The general equation for the refraction at a single spherical surface is given by the equation (6.56) is,

$$\frac{n_2}{v}-\frac{n_1}{u}=\frac{(n_2-n_1)}{R}$$

For the refracting surface ①, the light goes from $n_1$ to $n_2$.

$$\frac{n_2}{v'}-\frac{n_1}{u}=\frac{(n_2-n_1)}{R_1} \qquad (6.58)$$

For the refracting surface ②, the light goes from medium $n_2$ to $n_1$.

$$\frac{n_1}{v}-\frac{n_2}{v'}=\frac{(n_1-n_2)}{R_2} \qquad (6.59)$$

For surface ②, $I'$ acts as virtual object.

Adding the above two equations (6.58) and (6.59)

$$\frac{n_1}{v}-\frac{n_1}{u}=(n_2-n_1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

On further simplifying and rearranging,

$$\frac{1}{v}-\frac{1}{u}=\left(\frac{n_2-n_1}{n_1}\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

$$\frac{1}{v}-\frac{1}{u}=\left(\frac{n_2}{n_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right) \qquad (6.60)$$

If the object is at infinity, the image is formed at the focus of the lens. Thus, for $u=\infty$, $v=f$. Then the equation becomes.

$$\frac{1}{f}-\frac{1}{\infty}=\left(\frac{n_2}{n_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

$$\frac{1}{f}=\left(\frac{n_2}{n_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right) \qquad (6.61)$$

If the lens is kept in air, then we can take $n_1 = 1$ and $n_2 = n$. So the equation (6.61) becomes,

$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right) \qquad (6.62)$$

The above formula is called as the ***lens maker's formula***, because it tells the lens manufactures what curvature is needed for a material of particular refractive index to make a lens of desired focal length. This formula holds good also for any type of lens. By comparing the equations (6.60) and (6.61) we can write,

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \qquad (6.63)$$

The above equation is known as ***lens equation*** which relates the object distance $u$ and image distance $v$ with the focal length $f$ of the lens. This equation holds good for any type of lens.

## 6.6.4 Lateral magnification in thin lens

Let us consider an object $OO'$ of height $h_1$ placed on the principal axis with its height perpendicular to the principal axis. The inverted real image $II'$ is formed which has a height $h_2$ as shown in Figure 6.35.

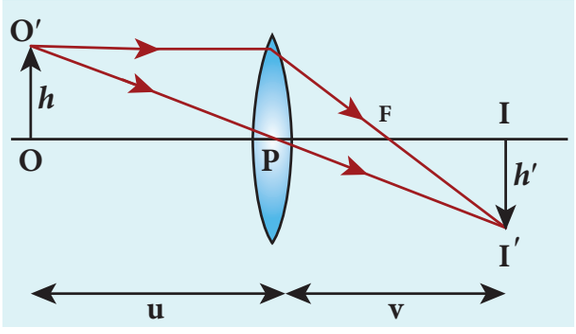



**Figure 6.35** Lateral magnification in thin lens

The lateral (or) transverse magnification $m$ is defined as the ratio of the height of the image to height of the object.

$$m=\frac{II'}{OO'} \qquad (6.64)$$

From the two similar triangles $\Delta POO'$ and $\Delta PII'$, we can write,

$$\frac{II'}{OO'}=\frac{PI}{PO} \qquad (6.65)$$



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